

Recording

Date	:	15/07/2026
Problem Specification	:	Check whether two arrays are equal.
Assumption	:	Both arrays contain valid integer values
Limitation	:	Both arrays must have the same length and the elements must appear in the same order
Input	:	Two integer arrays
Processing	:	Compare the lengths of the arrays. If the lengths are equal, compare each corresponding element one by one
Output	:	Display " This is equal " if both arrays are equal; otherwise display " This is not equal ".
Algorithm	:	Step 1: Create two integer arrays. Step 2: Check whether the lengths of both arrays are equal. Step 3: If the lengths are different, print " This is not equal " and stop. Step 4: Use a for loop to compare the corresponding elements of both arrays. Step 5: If any pair of elements is different, print " This is not equal " and stop. Step 6: If all elements are equal, print " This is equal ".
Programme listing	:	Programme file attached
Test data and expected output	:	1. Test data: int arr1[] = {1, 2, 3, 4, 5} int arr2[] = {1, 2, 3, 4, 5}; Expected output: This is equal 2. Test data: int arr1[] = {1, 2, 3, 4, 5}; int arr2[] = {1, 2, 2, 4, 5}; Expected output: This is not equal

Output obtained for test data :

1. Test data: `int arr1[] = {1, 2, 3, 4, 5};`
 `int arr2[] = {1, 2, 3, 4, 5};`
 Obtained Output: This is equal

2. Test data: `int arr1[] = {1, 2, 3, 4, 5}`
 `int arr2[] = {1, 2, 2, 4, 5};`

Obtained Output: This is not equal

Analysis : The numbers of operation required in performing the algorithm.

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Variable initialization	n		-	-
if condition	-	-	-	1
For loop	<i>N times</i>			<i>N</i>

Conclusion : The program correctly checks whether two arrays are equal by comparing their lengths and corresponding elements. It produces accurate results and has a time complexity of $O(n)$.

Discussion : The algorithm is simple and efficient for comparing two arrays.

 It compares each element in order and stops immediately if a mismatch is found, reducing unnecessary comparisons